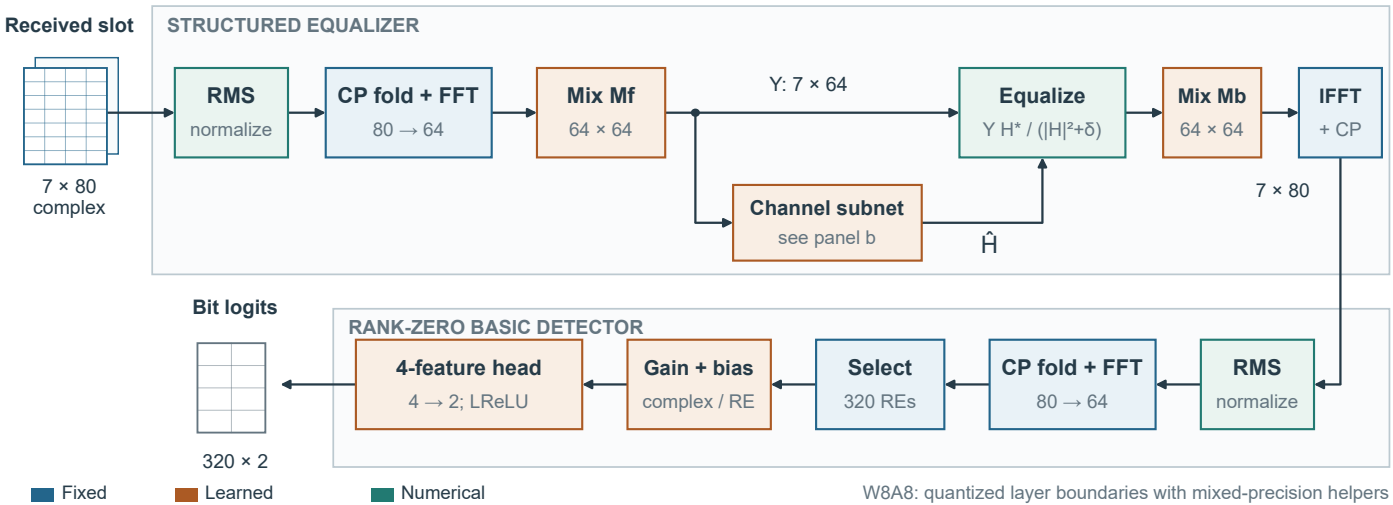
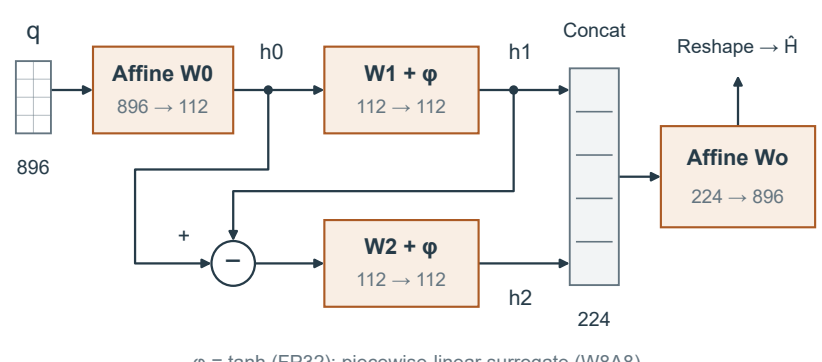
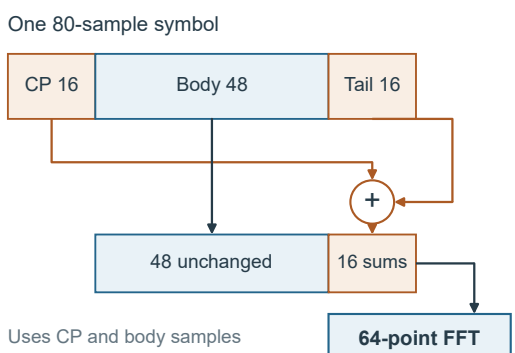

a Structured receiver: fixed Fourier backbone with learned corrections
Received slot
7 × 80 complex
STRUCTURED EQUALIZER
RMS
normalize
CP fold + FFT
80 → 64
Mix Mf
64 × 64
Y: 7 × 64
Channel subnet
see panel b
Ĥ
Equalize
Y H* / (|H|²+δ)
Mix Mb
64 × 64
IFFT
+ CP
7 × 80
Bit logits
320 × 2
RANK-ZERO BASIC DETECTOR
4-feature head
4 → 2; LReLU
Gain + bias
complex / RE
Select
320 REs
CP fold + FFT
80 → 64
RMS
normalize
Fixed
Learned
Numerical
W8A8: quantized layer boundaries with mixed-precision helpers


b Channel subnet: subtraction and feature reuse
q
896
Affine W0
896 → 112
h0
W1 + φ
112 → 112
h1
Concat
Reshape → Ĥ
Affine Wo
224 → 896
+
−
W2 + φ
112 → 112
h2
224
φ = tanh (FP32); piecewise-linear surrogate (W8A8)


c CP-aware Fourier operator
One 80-sample symbol
CP 16
Body 48
Tail 16
+
48 unchanged
16 sums
Uses CP and body samples
64-point FFT